HW 2

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(a) Given $Z_1, \ldots, Z_n$ are i.i.d. Bernoulli(p) for $p \leq 1/2$ and $t \in (0, 1-p)$, we can compare the upper bound

$$P\left[\frac{1}{n}\sum_{i=1}^{n} Z_i \geq t + p\right]$$

between Hoeffding's inequality, Bernstein's inequality, and Bennett's inequality. First, define $X_i = Z_i - p$, consequently, $E[Z_i] = p, Var(Z_i) = p(1-p)$, and thus $E[X_i] = 0, Var(X_i) = p(1-p)$. By this same definition, we can derive that

$$P\left(\frac{1}{n}\sum_{i=1}^{n} Z_i \geq t + p\right) = P\left(\sum_{i=1}^{n} X_i \geq nt\right)$$

Now for each inequality:

- **Hoeffding's inequality:** Hoeffding's inequality gives:

$$P\left(\sum_{i=1}^{n} X_i \geq s) \leq \exp(-\frac{2s^2}{n})\right), \quad s = nt \therefore$$
$$P\left(\frac{1}{n}\sum_{i=1}^{n} Z_i \geq t + p\right) \leq \exp(-2nt^2)$$

  thus, $R_H = 2nt^2$.

- **Bernstein's inequality:** Using X_i, since $Z_i \in \{0, 1\}$, $X_i \in \{-p, 1-p\}$. Because $p \leq 1/2$, $|X_i| \leq 1 - p$, and that $\sum_{i=1}^{n} Var(X_i) = np(1-p)$, Bernstein's inequality gives:

$$P\left(\sum_{i=1}^{n} X_i \geq s\right) \leq \exp\left(-\frac{s^2}{2(v + bs/3)}\right)$$

Again, let $s = nt$, with $v = np(1-p)$ and $b = 1-p$, we have:

$$\begin{aligned} P\left(\sum_{i=1}^{n} X_i \geq nt\right) &\leq \exp\left(-\frac{n^2t^2}{2np(1-p) + \frac{2}{3}(1-p)nt}\right) \\ &= \exp\left(-\frac{n^2t^2}{2n(1-p)(p+t/3)}\right) \\ &= \exp\left(-\frac{nt^2}{2(1-p)(p+t/3)}\right) \end{aligned}$$

thus, $R_{\text{Bern}} = \frac{nt^2}{2(1-p)(p+t/3)}$.

- **Bennett's inequality:** Using the same X_i as above, Bennett's inequality gives:

$$P\left(\sum_{i=1}^{n} X_i \geq s\right) \leq \exp\left(-\frac{v}{b^2}h\left(\frac{bs}{v}\right)\right), \text{ using } s, v, b \therefore$$

$$\begin{aligned} P\left(\sum_{i=1}^{n} X_i \geq nt\right) &\leq \exp\left(-\frac{np(1-p)}{(1-p)^2}h\left(\frac{(1-p)nt}{np(1-p)}\right)\right) \\ &= \exp\left(-\frac{np}{1-p}h\left(\frac{t}{p}\right)\right) \end{aligned}$$

thus, $R_{\text{Benn}} = \frac{np}{1-p}h\left(\frac{t}{p}\right)$.

(b) We can compute R_{Bern}/R_H using (a):

$$\frac{R_{\text{Bern}}}{R_H} = \frac{\frac{nt^2}{2(1-p)(p+t/3)}}{2nt^2} = \frac{1}{4(1-p)(p+t/3)}$$

We can see if Bernstein's bound improves over Hoeffding's more than a constant factor if $R_{\text{Bern}}/R_H \to \infty$ as $n \to \infty$. Since $p \leq \frac{1}{2}$, $\frac{1}{2} \leq 1-p \leq 1$, so $1-p$ is bounded away from 0. For the ratio to approach infinity, we need $p + t/3 \to 0$ as $n \to \infty$. This only occurs when $p \to 0$ and $t \to 0$ as $n \to \infty$. Thus, Bernstein's bound improves over Hoeffding's more than a constant factor when $p \to 0$ and $t \to 0$ as $n \to \infty$.

(c) $R_{\text{Benn}}/R_{\text{Bern}}$ depends on (p, t) only through $x = t/p$:

$$\begin{aligned} \frac{R_{\text{Benn}}}{R_{\text{Bern}}} &= \frac{\frac{np}{1-p}h\left(\frac{t}{p}\right)}{\frac{nt^2}{2(1-p)(p+t/3)}} = \frac{2p(p+t/3)}{t^2}h\left(\frac{t}{p}\right) \\ &= \frac{2p(p+px/3)}{p^2x^2}h(x) = \frac{2(1+x/3)}{x^2}h(x) \end{aligned}$$

Bennett's bound is never than Bernstein's since this ratio is always greater than 1:

$$\begin{aligned}
\frac{2(1+x/3)}{x^2}h(x) &\geq 1 \\
h(x) &\geq \frac{x^2}{2(1+x/3)} \\
\text{Let } f(x) &= h(x) - \frac{x^2}{2(1+x/3)} \\
h(x) &= (1+x)\log(1+x) - x \therefore \\
h'(x) &= \log(1+x), h''(x) = \frac{1}{1+x} \therefore \\
f''(x) &= \frac{1}{1+x} - \frac{1}{(1+x/3)^3} \geq 0
\end{aligned}$$

For all $x \geq 0$, $(1+x/3)^3 \geq 1+x$, so $f''(x) \geq 0$, then $f'(x) \geq 0$ also. With $f(0) = 0, f'(0) = 0$, we have $f(x) \geq 0$ for all $x \geq 0$. Therefore, $h(x) \geq \frac{x^2}{2(1+x/3)}$, and substituting this into the ratio gives:

$$\frac{2(1+x/3)}{x^2}h(x) \geq \frac{2(1+x/3)}{x^2} \cdot \frac{x^2}{2(1+x/3)} = 1$$

thus, Bennett's bound is never worse than Bernstein's.
As $x \to 0$, the taylor expansion of $h(x)$ gives $h(x) = \frac{x^2}{2} - \frac{x^3}{6} + O(x^4)$, so

$$\frac{2(1+x/3)}{x^2}h(x) = \frac{2(1+x/3)}{x^2}\left(\frac{x^2}{2} - \frac{x^3}{6} + O(x^4)\right) = 1 + \frac{x^2}{18} + O(x)$$

Thus, as $x \to 0$, the ratio approaches 1, and Bennett's bound is asymptotically equivalent to Bernstein's.
As $x \to \infty$, the taylor expansion of $h(x)$ gives $h(x) = x\log(x) - x + O(\log(x)) \sim x\log(x)$, so

$$\frac{2(1+x/3)}{x^2}h(x) = \frac{2(1+x/3)}{x^2}(x\log(x)) \sim \frac{2}{3}\log(x) = \infty$$

Thus, as $x \to \infty$, the ratio approaches infinity, and Bennett's bound is asymptotically better than Bernstein's.

2

We are given the ψ_α Orlicz norm of a random variable X defined as:

$$||X||_{\psi_\alpha} = \inf\left\{K > 0 : E\left[\psi_\alpha\left(\frac{|X|}{K}\right)\right] \leq 1\right\}$$

where $\alpha > 1$ and $\psi_\alpha(x) = e^{x^\alpha} - 1$. Using this, the Orlicz norm can be rewritten as:

$$||X||_{\psi_\alpha} = \inf\left\{K > 0 : E\left[e^{|X|^\alpha/K^\alpha}\right] \leq 2\right\}$$

Suppose $||X||_{\psi_\alpha} < \infty$. By the definition of the infimum, there exists a finite $K > 0$ where $E[e^{|X|^\alpha/K^\alpha}] \leq 2$. Then, for any $t > 0$, using Markov's inequality to the nonnegative random variable, we get that $Y = e^{|X|^\alpha/K^\alpha}$. This gives:

$$\begin{aligned} P(|X| \geq t) &= P(e^{|X|^\alpha/K^\alpha} \geq e^{t^\alpha/K^\alpha}) \\ &\leq \frac{E[e^{|X|^\alpha/K^\alpha}]}{e^{t^\alpha/K^\alpha}} \\ &\leq 2e^{-t^\alpha/K^\alpha} \end{aligned}$$

Choosing $C = 2$ and $c = 1/K^\alpha$, we have that $P(|X| \geq t) \leq Ce^{-ct^\alpha}$. Thus, $||X||_{\psi_\alpha} < \infty$ implies that there exists a $C, c > 0$ such that $P(|X| \geq t) \leq Ce^{-ct^\alpha}$.

Now, we want to show that there exists a finite $K > 0$ such that $E[e^{|X|^\alpha/K^\alpha}] \leq 2$ if there exists a $C, c > 0$ such that $P(|X| \geq t) \leq Ce^{-ct^\alpha}$. Let $Y = |X|^\alpha \geq 0$, then for any $u \geq 0$:

$$\begin{aligned} P(Y \geq u) = P(|X|^\alpha \geq u) &= P(|X| \geq u^{1/\alpha}) \\ &\leq Ce^{-c(u^{1/\alpha})} = Ce^{-cu} \end{aligned}$$

For any $a > 0$, $e^{aY} = 1 + \int_0^Y ae^{au}du$. The tail integration identity for nonnegative random variables gives us that $E[e^{aY}] = 1 + \int_0^\infty ae^{au}P(Y \geq u)du$. Using the tail bound on $P(Y \geq u)$, we have:

$$\begin{aligned} E[e^{aY}] &\leq 1 + \int_0^\infty ae^{au}Ce^{-cu}du \\ &= 1 + aC\int_0^\infty e^{-(c-a)u}du \\ &= 1 + \frac{aC}{c-a} \end{aligned}$$

Then if we choose $a = \frac{c}{C+1}$, since $C, c > 0$, $0 < a < c$, and we have:

$$\begin{aligned} E[e^{aY}] &\leq 1 + \frac{C(\frac{c}{C+1})}{c - \frac{c}{C+1}} \\ &= 1 + \frac{\frac{Cc}{C+1}}{\frac{cC}{C+1}} \\ &= 1 + 1 = 2 \end{aligned}$$

Thus, $E[e^{aY}] \leq 2$. Recall that $Y = |X|^\alpha$, so we choose $K > 0$ such that $\frac{1}{K^\alpha} = a = \frac{c}{C+1}$, and solving for K gives us $K = \left(\frac{C+1}{c}\right)^{1/\alpha}$. Then we have that $E[e^{|X|^\alpha/K^\alpha}] \leq 2$, and then by the definition of the Orlicz norm, there exists a finite $K > 0$. Because both implications hold, we have that:

$$||X||_{\psi_\alpha} < \infty \iff \exists C, c > 0 : P(|X| \geq t) \leq Ce^{-ct^\alpha} \quad \forall t \geq 0$$

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From the definiton of the ψ_α Orlicz norm, we have that:

$$||X||_{\psi_\alpha} = \inf\left\{K > 0 : E\left[e^{|X|^\alpha/K^\alpha}\right] \leq 2\right\}$$

Define:

$$\beta = \frac{\alpha}{\alpha - 1} > 2 \therefore \frac{1}{\alpha} + \frac{1}{\beta} = 1, \beta(\alpha - 1) = \alpha$$

By taking the Taylor expansion of the MGF $M(\lambda) = E[e^{\lambda X i}]$, we have:

$$\begin{aligned}
e^{\lambda X_i} &\leq 1 + \lambda X_i + \frac{\lambda^2 |X_i|^2}{2} e^{|\lambda X_i|} \\
E[e^{\lambda X_i}] &\leq 1 + \lambda E[X_i] + \frac{\lambda^2}{2} E[|X_i|^2 e^{|\lambda X_i|}] \\
&\leq 1 + \frac{\lambda^2}{2} E[|X_i|^2 e^{|\lambda X_i|}], \lambda \in [0, 1] \therefore e^{\lambda X_i} \leq e^{|X_i|} \therefore \\
E[e^{\lambda X_i}] &\leq 1 + \frac{\lambda^2}{2} E[X_i^2 e^{|X_i|}]
\end{aligned}$$

We now need to bound $E[X_i^2 e^{|X_i|}]$. We want to find constant $C > 0$ such that $X_i^2 e^{|X_i|} \leq Ce^{|X_i|^\alpha/K^\alpha}$, rearrangeing gives:

$$X_i^2 e^{|X_i| - |X_i|^\alpha/K^\alpha} \leq C$$

Because $\alpha > 1$, we have that $|X_i| - |X_i|^\alpha/K^\alpha \to -\infty$ as $|X_i| \to \infty$, then $X_i^2 e^{|X_i| - |X_i|^\alpha/K^\alpha} \to 0$ as $|X_i| \to \infty$. Then, there exists some constant $C > 0$ such that $X_i^2 e^{|X_i| - |X_i|^\alpha/K^\alpha} \leq C$ for all $|X_i| \geq 0$. Then, we have that:

$$\begin{aligned}
E[X_i^2 e^{|X_i|}] &\leq CE[e^{|X_i|^\alpha/K^\alpha}] \\
&\leq 2C, \therefore \\
E[e^{\lambda X_i}] &\leq 1 + \frac{\lambda^2}{2} 2C \leq e^{C\lambda^2}, \lambda \in [0, 1] \\
\log E[e^{\lambda X_i}] &\leq C\lambda^2, \lambda \in [0, 1]
\end{aligned}$$

Now we bound the MGF for $\lambda > 1$. Using Young's inequality, we have:

$$ab \leq \frac{a^\alpha}{\alpha} + \frac{b^\beta}{\beta}$$

Choosing $a = \frac{\alpha^{1/\alpha}|X_i|}{K}$ and $b = \frac{\lambda K}{\alpha^{1/\alpha}}$, we have:

$$ab = \lambda|X_i|$$
$$\lambda|X_i| \leq \frac{|X_i|^\alpha}{K^\alpha} + \frac{\lambda^\beta K^\beta}{\beta\alpha^{\beta/\alpha}}$$

Let $C = \frac{K^\beta}{\beta\alpha^{\beta/\alpha}}$, then we have:

$$\begin{aligned}
\lambda|X_i| &\leq \frac{|X_i|^\alpha}{K^\alpha} + C\lambda^\beta \\
e^{\lambda|X_i|} &\leq e^{|X_i|^\alpha/K^\alpha} e^{C\lambda^\beta} \\
E[e^{\lambda|X_i|}] &\leq E[e^{|X_i|^\alpha/K^\alpha}]e^{C\lambda^\beta} \\
&\leq 2e^{C\lambda^\beta}, \text{ collapse constants } \therefore \\
\log E[e^{\lambda|X_i|}] &\leq (\log 2 + C)\lambda^\beta, \lambda > 1, \beta = \frac{\alpha}{\alpha - 1}, \therefore \\
\log E[e^{\lambda X_i}] &\leq C\lambda^{\alpha/(\alpha-1)}, \lambda > 1
\end{aligned}$$

Now, we can combine the two bounds on the MGF for $\lambda \in [0, 1]$ and $\lambda > 1$, and establish some constant $C > 0$ that depends only on K, α, such that:

$$\log E[e^{\lambda X_i}] \leq C(\lambda^2 + \lambda^\beta) \forall \lambda \geq 0$$

Let $S_n = \sum_{i=1}^n X_i$, then by Markov's inequality, and X_i independence, we have:

$$\begin{aligned}
P\left(\frac{S_n}{n} \geq t\right) &= P\left(e^{\lambda S_n} \geq e^{\lambda nt}\right) \\
&\leq \frac{E[e^{\lambda S_n}]}{e^{\lambda nt}} \\
\log E[e^{\lambda S_n}] = \log E[e^{\lambda \sum_{i=1}^n X_i}] &= \log \prod_{i=1}^n E[e^{\lambda X_i}] = \sum_{i=1}^n \log E[e^{\lambda X_i}] \\
&\leq \sum_{i=1}^n C(\lambda^2 + \lambda^\beta) = nC(\lambda^2 + \lambda^\beta)
\end{aligned}$$

Thus, we have:

$$P\left(\frac{S_n}{n} \geq t\right) \leq \exp(nC(\lambda^2 + \lambda^\beta) - \lambda nt)$$

- $0 < t \leq 1$**:** Choose some $\lambda = \delta t$, where $\delta > 0$ and is sufficiently small depending only on C, β. Then we have:

$$\begin{aligned} P\left(\frac{S_n}{n} \geq t\right) &\leq \exp(nC(\delta^2 t^2 + \delta^\beta t^\beta) - \delta n t^2) \\ &= \exp(n(C\delta^2 - \delta)t^2 + nC\delta^\beta t^\beta) \\ &\beta > 2, 0 < t \leq 1, \therefore t^\beta \leq t^2, \\ &\leq \exp(n(\delta^2 - C\delta^2 - C\delta^\beta)t^2), \text{ collapse constants} \\ &\leq \exp(-c_1 n t^2), \text{ for some } c_1 > 0, \text{ sufficiently small } \delta \end{aligned}$$

- $t > 1$**:** Choose some $\lambda = \delta t^{\alpha-1}$, again with sufficiently small δ:

$$\begin{aligned} P\left(\frac{S_n}{n} \geq t\right) &\leq \exp(nC(\delta^2 t^{2(\alpha-1)} + \delta^\beta t^{\beta(\alpha-1)}) - \delta n t^\alpha) \\ &= \exp(n(C\delta^2 t^{2(\alpha-1)} + C\delta^\beta t^\alpha) - \delta n t^\alpha) \\ &= \exp(n(C\delta^2 t^{2(\alpha-1)} + (C\delta^\beta - \delta)t^\alpha)) \\ &t > 1, 2(\alpha-1) < \alpha, \therefore t^{2(\alpha-1)} < t^\alpha, \\ &\leq \exp(n(C\delta^2 + C\delta^\beta - \delta)t^\alpha), \text{ collapse constants} \\ &\leq \exp(-c_2 n t^\alpha), \text{ for some } c_2 > 0, \text{ sufficiently small } \delta \end{aligned}$$

Now combine both cases, and we have that for some constants $c_1, c_2 > 0$:

- **For** $0 < t \leq 1$**:** $\min(t^2, t^\alpha) = t^2$
- **For** $t > 1$**:** $\min(t^2, t^\alpha) = t^\alpha$

$$\begin{aligned} P\left(\frac{1}{n}\sum_{i=1}^n X_i \geq t\right) &\leq \begin{cases} e^{-cnt^2}, & 0 < t \leq 1 \\ e^{-cnt^\alpha}, & t > 1 \end{cases} \\ P\left(\frac{1}{n}\sum_{i=1}^n X_i \geq t\right) &\leq C\exp\left(-cn\min(t^2, t^\alpha)\right), \quad \forall t > 0 \end{aligned}$$

AI Usage: All homework problems were written in VSCode and compiled using LaTeX, with assistance from ChatGPT and Github Copilot inline suggestions.